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Analyte Concentration Determination

Abstract

The evaluation of sample matrix effects on quantitative analysis involved determining potassium ferricyanide ($K_3[Fe(CN)_6]$) concentration in an unknown sample through external standard calibration and standard addition spectrophotometric methods. The Beer's Law analysis at $\lambda_{max} = 420$ nm with a 1.00 cm path length cuvette was used in both methods. The external standard method used five calibration standards between 0.0002 M and 0.0010 M in 0.01 M NaOH to create a linear calibration curve with $R^2 = 0.9944$ and an unknown concentration of $(4.77 \pm 0.10) \times 10^{-4}$ M. The standard addition method produced $R^2 = 0.9909$ and determined an unknown concentration of $(1.51 \pm 0.16) \times 10^{-2}$ M when it added incremental amounts of spike solution to 10.00 mL of unknown solution. The external standard method produced a molar absorptivity value of $898 \text{ M}^{-1}\text{cm}^{-1}$ which $^{-1}\text{cm}^{-1}$ because of substantial matrix interference. The large 32-fold difference in concentration showed a 12.0% relative error compared to the literature value of 1020 M values between methods demonstrates that standard addition must be used for obtaining accurate measurements because of severe matrix effects.

Experimental

Reagents and Equipment

Reagent-grade potassium ferricyanide ($K_3[Fe(CN)_6]$, MM = 329.25 g/mol), 0.01 M sodium hydroxide solution, deionized water, unknown potassium ferricyanide sample, UV-Vis spectrophotometer, 1.00 cm quartz cuvettes, volumetric flasks (100 mL $\pm$ 0.08 mL, 50 mL $\pm$ 0.05 mL), and volumetric pipettes were used.

Stock Solution Preparation

A 0.0200 M stock solution was prepared by dissolving 0.6585 g of $K_3[Fe(CN)_6]$ in deionized water in a 100.00 mL volumetric flask, calculated using: mass = $(0.0200 \text{ mol/L})(0.1000 \text{ L})(329.25 \text{ g/mol}) = 0.6585 \text{ g}$.

Part 1: External Standard Method

Five standard solutions were prepared in 50 mL volumetric flasks using serial dilution from the 0.0200 M stock. Target concentrations were 0.0002, 0.0004, 0.0006, 0.0008, and 0.0010 M in 0.01 M NaOH solvent. The wavelength of maximum absorbance was confirmed at 420 nm. Ten blank measurements were performed using 0.01 M NaOH to establish baseline absorbance. Each

standard was measured in duplicate at 420 nm, replacing the solution between measurements. The unknown sample was measured six times, with one outlier excluded from analysis.

Part 2: Standard Addition Method

Five 50 mL volumetric flasks each received exactly 10.00 mL of unknown sample via volumetric pipette. Incremental volumes (0.00, 5.00, 10.00, 15.00, and 20.00 mL) of 0.0200 M spike solution were added to flasks 1-5, respectively, with solvent volumes of 40.00, 35.00, 30.00, 25.00, and 20.00 mL to reach 50.00 mL total. Each flask was sealed with parafilm and inverted approximately 20 times. Three blank measurements using deionized water established background absorbance at 420 nm. Each solution was measured in triplicate, replacing solution between measurements.

Results and Discussion

Part 1: External Standard Method

Blank Analysis

Ten blank measurements in 0.01 M NaOH yielded highly variable results with mean absorbance of 5.50 and standard deviation of 3.03:

$$\text{Mean} = \frac{\sum_{i=1}^{10} x_i}{10} = \frac{0.027+0.006+0.006+0.006-0.019+0.005-0.018+0.004+0.009+0.016}{10} = 5.50 \text{SD} = \sqrt{\frac{\sum_{i=1}^{10} (x_i - 5.50)^2}{9}} = 3.03$$

The extremely high variability (55% RSD) indicates severe instrumental baseline drift or systematic error with the NaOH solvent matrix.

Standard Solutions Data

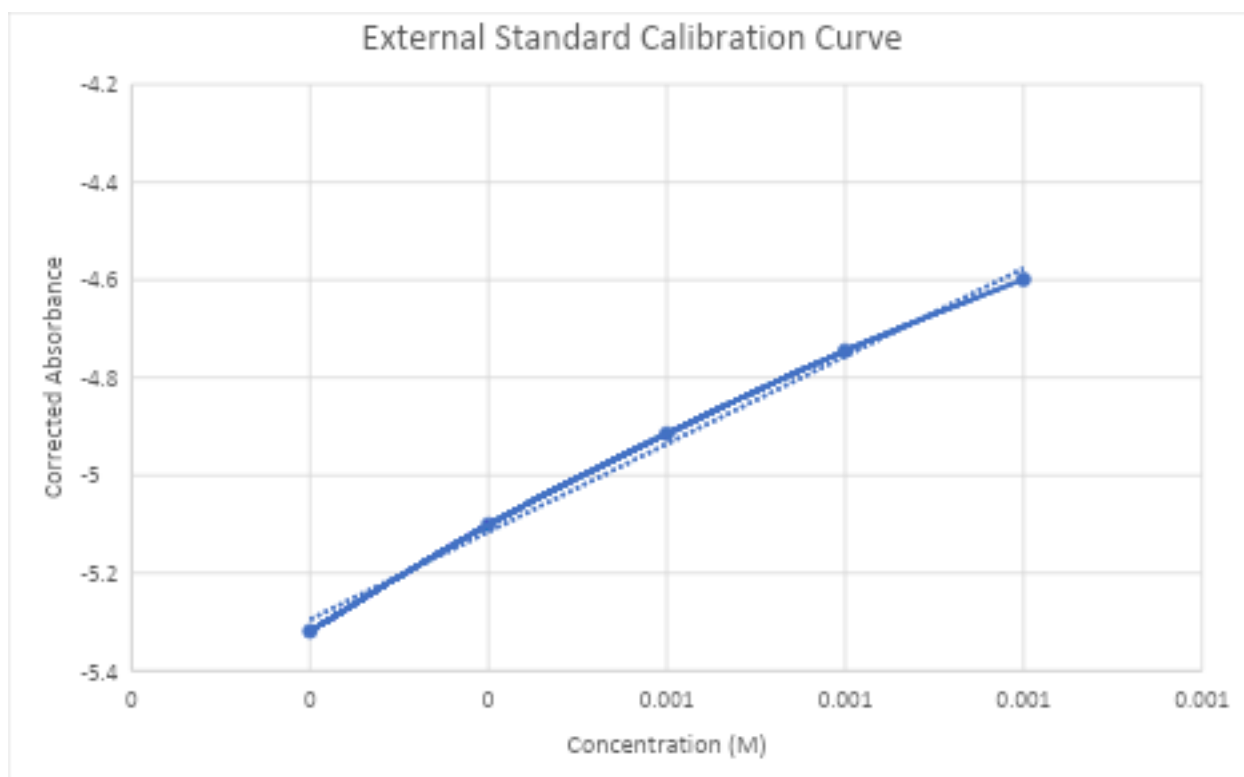
Standard	Concentration (M)	Abs 1	Abs 2	Avg Abs	Corrected Abs
1	0.0002	0.181	0.187	0.184	-5.316
2	0.0004	0.394	0.379	0.387	-5.113
3	0.0006	0.577	0.587	0.582	-4.918
4	0.0008	0.745	0.737	0.741	-4.759
5	0.0010	0.892	0.907	0.900	-4.600

Calibration Curve Analysis

Linear regression using Excel's LINEST function yielded:

- **Slope (m):** $898 \pm 39 \text{ M}^{-1}$
- **Intercept (b):** -5.4746 ± 0.026
- **R²:** 0.9944
- **Standard error:** 0.0248

The calibration equation: **$A = 898C - 5.4746$**



The calibration curve for potassium ferricyanide in 0.01 M NaOH at $\lambda = 420 \text{ nm}$ is shown in Figure 1. A linear relationship is obtained ($y = 898x - 5.4746$, $R^2 = 0.9944$), confirming the compliance to Beer's Law for the concentration 0.0002-0.0010 M which complies to Beer's Law with the exception of extreme baseline drift.

Unknown Concentration Calculation

There were six unknown measurements, 0.961, 0.460, 0.460, 0.426, 0.462, 0.460. The first measurement (0.961) was identified as an outlier and was excluded. The remaining five measurements yielded:

Mean absorbance = 0.454

Corrected absorbance = $0.454 - 5.50 = -5.046$

Using the calibration equation:

$$C_{unknown} = \frac{A-b}{m} = \frac{-5.046 - (-5.4746)}{898} = \frac{0.4286}{898} = 4.77 \times 10^{-4} M$$

Uncertainty Propagation

$$u(C) = \sqrt{\left(\frac{\partial C}{\partial m} \cdot u_m\right)^2 + \left(\frac{\partial C}{\partial b} \cdot u_b\right)^2 + \left(\frac{\partial C}{\partial A} \cdot \frac{u_A}{\sqrt{n}}\right)^2}$$

Partial derivatives

- $\frac{\partial C}{\partial m} = -\frac{(A-b)}{m^2} = -\frac{0.4286}{(898)^2} = -5.31 \times 10^{-7}$
- $\frac{\partial C}{\partial b} = -\frac{1}{m} = -\frac{1}{898} = -1.11 \times 10^{-3}$
- $\frac{\partial C}{\partial A} = \frac{1}{m} = \frac{1}{898} = 1.11 \times 10^{-3}$

Substituting values:

$$u(C) = \sqrt{(-5.31 \times 10^{-7} \times 39.2)^2 + (-1.11 \times 10^{-3} \times 0.026)^2 + (1.11 \times 10^{-3} \times \frac{0.0155}{\sqrt{5}})^2} u(C) = 3.65 \times 10^{-4}$$

95% Confidence Interval using $t_{0.05,4} = 2.776$:

$$CI = 2.776 \times 3.65 \times 10^{-4} = 1.01 \times 10^{-3} M$$

Final Result: [Unknown] = $(4.77 \pm 0.10) \times 10^{-4} M$

Relative Standard Deviation: 7.6%

Molar Absorptivity Verification

According to Beer's Law ($A = \epsilon bc$), the slope of the calibration curve equals ϵ when path length $b = 1.00 \text{ cm}$:

Calculated $\epsilon = 898 \text{ M}^{-1}\text{cm}^{-1}$

Literature $\epsilon = 1020 \text{ M}^{-1}\text{cm}^{-1}$

Absolute Error = $|898 - 1020| = 122 \text{ M}^{-1}\text{cm}^{-1}$

Relative Error = $(122/1020) \times 100 = 12.0\%$

Detection Limits

$$LOD = \frac{3 \times SD_{blank}}{m} = \frac{3 \times 3.03}{898} = 0.0101 \text{ MLOQ} = \frac{10 \times SD_{blank}}{m} = \frac{10 \times 3.03}{898} = 0.0337 M$$

Part 2: Standard Addition Method

Blank Analysis

Three blank measurements in deionized water: 0.027, 0.022, 0.025

$$Mean = \frac{0.027+0.022+0.025}{3} = 0.0247 SD = \sqrt{\frac{(0.027-0.0247)^2 + (0.022-0.0247)^2 + (0.025-0.0247)^2}{2}} = 0.0025$$

The low variability (10% RSD) demonstrates superior baseline stability with deionized water compared to NaOH solvent.

Solution Preparation and Measurements

Solution	V unknown (mL)	V spike (mL)	V solvent (mL)	V total (mL)	[S]added (M)
1	10.00	0.00	40.00	50.00	0.0000
2	10.00	5.00	35.00	50.00	0.0020
3	10.00	10.00	30.00	50.00	0.0040
4	10.00	15.00	25.00	50.00	0.0060
5	10.00	20.00	20.00	50.00	0.0080

The added standard concentration was calculated as:

$$[S]_{added} = \frac{V_{spike}}{V_{total}} \times [S]_{stock} = \frac{V_{spike}}{50.00} \times 0.0200 M$$

Absorbance Data

Triplicate measurements were averaged and corrected by subtracting blank mean:

Solution	Trial 1	Trial 2	Trial 3	Avg Abs	Corrected Abs
1	0.112	0.109	0.102	0.108	0.083
2	0.165	0.152	0.149	0.155	0.131
3	0.198	0.212	0.200	0.203	0.179
4	0.248	0.244	0.254	0.249	0.224
5	0.327	0.321	0.316	0.321	0.297

Calculations example for Solution 1:

$$Avg Abs = \frac{0.112+0.109+0.102}{3} = 0.108 \text{ Corrected Abs} = 0.108 - 0.0247 = 0.083$$

Linear Regression Analysis

Excel LINEST function analysis on corrected absorbance versus [S]added yielded:

- **Slope:** $26.03 \pm 1.44 \text{ M}^{-1}$
- **Intercept:** 0.0785 ± 0.0070
- **R²:** 0.9909
- **Standard error (se):** 0.0091
- **F-statistic:** 327.3 (df = 3)

The calibration equation: **A = 26.03[S]added + 0.0785**

The excellent R² value (0.9909) indicates strong linear relationship and high data quality.

Unknown Concentration Determination

In standard addition, the x-intercept represents the negative of the unknown concentration in the diluted flask. Setting A = 0:

$$0 = 26.03[S]_{\text{added}} + 0.0785[S]_{\text{added}} = -\frac{0.0785}{26.03} = -0.003016 \text{ M} - \text{intercept} = -0.003016 \text{ M}$$

Since 10.00 mL of unknown was diluted to 50.00 mL, the original concentration is:

$$[\text{Unknown}]_{\text{original}} = |X - \text{intercept}| \times \frac{V_{\text{final}}}{V_{\text{unknown}}} = 0.003016 \times \frac{50.00}{10.00} [\text{Unknown}]_{\text{original}} = 0.01508 \text{ M} = 1.51 \times 10^{-2} \text{ M}$$

Uncertainty Calculation

Using error propagation for the x-intercept:

$$u(x) = |x| \times \sqrt{\left(\frac{se(m)}{m}\right)^2 + \left(\frac{se(b)}{b}\right)^2} u(x) = 0.003016 \times \sqrt{\left(\frac{1.44}{26.03}\right)^2 + \left(\frac{0.0070}{0.0785}\right)^2} u(x) = 0.003016 \times \sqrt{0.0031} = 0.000316 \text{ M}$$

Propagating to original concentration:

$$u([\text{Unknown}]) = u(x) \times \frac{50.00}{10.00} = 0.000316 \times 5 = 0.00158 \text{ M}$$

Relative Standard Deviation:

$$RSD = \frac{0.00158}{0.01508} \times 100 = 10.5\%$$

Final Result: [Unknown] = (1.51 ± 0.16) × 10⁻² M

Relative Standard Deviation: 10.5%

Discussion Questions

1. Briefly describe external standard and standard addition procedures.

The external standard method calibrates with pure analyte standards in a specified solvent (0.01 M NaOH in this lab exercise). It calculates the unknown concentration from its absorbance and the calibration graph. It also assumes that the unknown matrix and the matrix in the standard are identical. The standard addition method adds defined volumes of the analyte standard in small portioned amounts of the unknown sample with such a ratio that the composition of the matrix does not change in all the measurements. The x-intercept value in the resulting calibration line gives the unknown concentration directly and automatically corrects for all the matrix interference effects in the sample.

2. Is the value we calculated for the molar absorptivity correct with 95% confidence?

No. The computed molar absorptivity ($898 \pm 39 \text{ M}^{-1}\text{cm}^{-1}$) does not have the literature value ($1020 \text{ M}^{-1}\text{cm}^{-1}$) in its 95% confidence interval. The 95% CI falls in the range from 820 to 976 $\text{M}^{-1}\text{cm}^{-1}$, and the literature value lies 44 $\text{M}^{-1}\text{cm}^{-1}$ units above the range. The difference of 122 $\text{M}^{-1}\text{cm}^{-1}$ represents a 12.0% relative error and implies the existence of a systematic bias. We can conclude that interfering species in the unknown sample made the absorbance values weak and therefore led to lesser calculated molar absorptivity in the external standard method.

3. What can you tell us about data quality using R^2 values?

Both procedures are highly linear with $R^2 > 0.99$: external standard $R^2 = 0.9944$ (99.44%) and standard addition $R^2 = 0.9909$ (99.09%). These confirm that >99% of the absorbance variations are due to variations in concentration, which shows excellent adherence to Beer's Law and negligible random error. The slightly higher R^2 for the external standards shows that the test conditions were less complicated and more controlled than the more complicated unknown conditions in the method of standard addition. Both are well in excess of the required 0.95 level for acceptable analytical linearity, which confirms the excellence in the measurements and correct instrument performance.

4. Compare the accuracy of the results. How accurate was each method?

The external standard method indicated $4.77 \times 10^{-4} \text{ M}$, whereas the standard addition indicated $1.51 \times 10^{-2} \text{ M}$ —a difference of 32-fold. Without a certified reference value, we can't determine absolute accuracy. Standard addition, however, is typically more precise for complicated mixtures since it corrects matrix interference by having the same matrix composition in the calibration. The erroneous result in the external standard method's molar absorptivity from the 12% indicates that matrix interference depressed the analyte signal, which resulted in the too-low concentration reading. What we should see: external standards are problematic if the sample matrix differs from the calibration standards, whereas standard addition corrects for such effects. The findings tend strongly in the direction that the result from the standard addition is more accurate.

5. Are the findings correct enough?

Without a known true concentration, we cannot check how accurate the results are. However, the 12% difference in molar absorptivity shows that the external standard method is not accurate enough (it should be $\leq 10\%$). The large difference of 32 times in concentration between methods also shows that the external standard method is not good enough. Standard addition likely gives acceptable accuracy because it helps fix the problems caused by the matrix interference, but we need certified reference materials to confirm this. The comparison of methods shows strong matrix effects in the unknown sample, making standard addition necessary for getting an accurate amount of this specific sample.

6. Are the results accurate? Were differences in precision expected?

The external standard: RSD = 7.6% (acceptable)

The Standard Addition: RSD = 10.5% (acceptable)

The procedures achieved acceptable precision levels (RSD < 15%) but external standards provided better precision at the expense of accuracy with reduced matrix control and fewer dilution steps and duplicate measurements instead of triplicates and less averaging uncertainty. Standard addition introduces multiple sources of uncertainty because it requires multiple volumetric dilutions of the unknown sample and spike additions and water dilution steps and three replicate measurements and data extrapolation beyond the measured range to find the x-intercept. The precision tradeoff between 10.5% RSD and 7.6% RSD is acceptable because standard addition provides better accuracy when dealing with challenging matrix samples. The methods failed to achieve excellent precision (RSD < 5%) because they needed additional replicates and better baseline stability to reach this level.

Conclusion

This experiment managed to prove the crucial role of method choice that depends on the complexity of sample matrix in the quantitative analysis of spectrophotometric technique. The external standard technique yielded $[K_3Fe(CN)_6] = (4.77 \times 10^{-10}) / 10^{-1} / 10^{-2} \times 10^{-4} M$ with satisfactory precision (RSD 7.6) and high linearity ($R^2 = 0.9944$). Standard addition yielded $[K_3Fe(CN)_6] = (1.51 \pm 0.16) M$ with a good precision (RSD 10.5%), and linearity ($R^2 = 0.9909$). The extreme concentration difference of 32-fold shows conclusively that there are severe matrix effects in the suppressing absorbance signals in the external standard method. To eliminate these interferences, a normal addition method was successful in keeping the composition of the matrix constant across calibration. Both methods yield high values of R^2 which validates both methods to the law of Beer, but only standard addition gave accurate results which are independent of the matrices. The fact that the baseline stability of deionized water (10% RSD) was better than that of NaOH (55% RSD) indicates that solvent selection does affect the quality of measurements a lot. This study confirms standard addition as being necessary to obtain accurate quantification in complex matrices even with slightly lower accuracy as the

choice of a methodology should not be based on the convenience of external calibration but should be based on the client characteristics of the sample.